

The Second Workshop on Intelligent and Interactive Writing Assistants

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ABSTRACT

In recent years, writing assistants have become increasingly sophisticated and ubiquitous, fueled by advances in artificial intelligence, particularly large language models. As new use cases and models emerge, we expect the adoption rate to accelerate. In this interdisciplinary workshop, we, as a diverse group of researchers and practitioners interested in intelligent and interactive writing assistants, will create a taxonomy of writing assistants and discuss their desirable features and potential consequences. We invite writers, educators, researchers, industry practitioners, students, and anyone interested in creating, using, and testing future writing assistant technologies to join the conversation.

CCS CONCEPTS

• **Human-centered computing**; • **Computing methodologies**
→ **Natural language processing**

KEYWORDS

Writing assistants, Writing support tools, Creativity support tools, AI-assisted writing, Human-AI interaction, Language models, Human-computer interaction, Natural language processing

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1 BACKGROUND

Writing is a critical skill in daily life and professional environments. It provides means for communicating with family, friends, and colleagues, sharing information, as well as enhancing learning and emotional well-being [1, 2, 5, 6]. Writing also underlies a wealth of cultural activity from poetry to political commentary. However, writing is oftentimes a challenging endeavor. From language learners to journalists, almost everyone struggles in some way with the translation of amorphous ideas in our mind into coherent words on the page. For instance, writers often struggle to find appropriate phrasing, set the right tone, organize their thoughts logically, and in general, find ways to convey their thoughts clearly to be properly understood.

Nowadays, many writing assistants are seamlessly integrated into our personal devices (e.g., cell phones and laptops) and software (e.g., word processors and email software) to support our writing process.¹ Some writing assistants (e.g., Grammarly and Wordtune) can be installed as browser extensions and used to write and revise in any web page. Beyond spelling and grammar checking, we have seen an enormous shift in the ability of writing assistants from paraphrasing to auto-completion to content generation² in the past few years. Intelligent writing assistants have also been used to co-author multiple books and award-winning texts.³

However, the design and development of writing assistants still pose many challenges. Their wide range of assistance is fueled by recent advances in the field of artificial intelligence (AI), particularly large language models [3, 4, 8]. As a result, when building writing assistants, various interdisciplinary design decisions have to be made, such as the way of supporting user goals and needs, the choice of the backend language model(s) and their adaptation methods, the granularity level of writing feedback (e.g., analyzing

¹Examples of autocomplete systems: Apple iOS’s predictive text, Microsoft Word’s text predictions, Google Gmail’s smart compose

²Examples of commercial tools for copywriting: Copy.ai, Copysmith, Jarvis; for creative writing: AI Dungeon, Sudowrite Rytr, QuillBot, Writesonic

³Examples of texts co-authored with AI: Ghosts (Best American 2022 Selection), Nothing Breaks Like A.I. Heart (An essay about artificial intelligence, emotional intelligence, and finding an ending)

words or sentences), the way to handle potential failures from AI, and the display of suggestions to users (e.g., in-text highlighting vs. dashboard visualizations). While scholars have started to assess the effects of writing feedback on users' task outcomes (e.g., general essay writing [7], argumentation [9], or writing styles), our understanding is yet limited regarding the granular effects of different writing support modalities (e.g., highlighting, dashboards, or autocomplete) as well as the long-term effects of such support on users' cognitive abilities, their long-term writing styles, and their psychological effects.

Therefore, in our workshop at CHI 2023, we aim to facilitate discussion around the following questions:

- What are important dimensions to consider when designing intelligent writing assistants?
- How can we evaluate and compare writing assistants and their features in interactive settings?
- How can we design writing assistants for underrepresented languages, types of writers, and writing tasks?
- How can we make writing assistants more accessible and inclusive?
- What are the biggest ethical concerns and limitations of writing assistants, especially when leveraging AI?
- How do we address legal issues with copyright and the psychological sense of ownership of writing assistants?
- How will writing assistants change the way we write?

We believe our collective effort to answer these questions can inform researchers to build and evaluate human-centered writing assistants. To this end, we strive to bring together researchers from the human-computer interaction (HCI) and natural language processing (NLP) communities by alternating our workshop venue between HCI and NLP every year.

2 ORGANIZERS

Minsuk Chang (Google) is a research scientist on the PAIR (People and AI Research) team at Google Research. He explores novel computational interaction techniques powered by AI technologies by designing, building, and evaluating AI powered systems with users in the loop. Specifically, he is interested in 1) making large-scale pretrained models usable and useful, and 2) leveraging these pretrained models for simulating users and interfaces. He is also an adjunct professor at the KAIST Graduate School of AI.

John Joon Young Chung (University of Michigan) is a Ph.D. candidate at the University of Michigan advised by Eytan Adar. His research topic is in studying, designing, implementing, and evaluating creativity support tools for art-making by leveraging AI/ML technologies. In the domain of visual arts and creative writing, he built tools that turn end-to-end generative models into more interactive and iterative systems that align with the progressive human creation process.

Katy Ilonka Gero (Columbia University) is a Ph.D. candidate at Columbia University in the City of New York. Her research focuses on designing and studying new paradigms for writing support using NLG, and how we can mitigate the problems associated with large models, like the generation of offensive language. She is on the organizing committee for the Workshop on Human-AI Co-Creation

with Generative Models, held annually for the past three years at the Intelligent User Interfaces conference.⁴

Ting-Hao 'Kenneth' Huang (Pennsylvania State University) is an Assistant Professor at the College of Information Sciences and Technology (IST) of the Pennsylvania State University. His research lies in the intersection of Artificial Intelligence (AI) and Human-Computer Interaction (HCI), imagining new possibilities of human-AI collaborations. Dr. Huang explores the creative and complex domains, such as open conversation, writing support, and automatic storytelling, which seem exceptionally challenging to automate. His work aims to move automation beyond low-level, mundane tasks to augment human creativity and sociability.

Dongyeop Kang (University of Minnesota) is an Assistant Professor in the Computer Science Engineering department at the University of Minnesota, Twin Cities. He leads the Minnesota Natural Language Processing (NLP) group that aims to develop human-centered language technologies, focusing on developing interdisciplinary methods for NLP models and building interactive NLP systems for scientists and creative writers. He is co-organizing the Controllable Generative Modeling of Language and Vision (Ctrl-Gen) workshop⁵ at NeurIPS 2021.

Mina Lee (Stanford University) is a final-year Ph.D. candidate at Stanford University, who leverages language models to build writing assistants, including an autocomplete system, a contextual thesaurus system, and a general-purpose text editor with suggestions. She identifies and measures how these models change the way we write or enhance our productivity and creativity. For instance, they can help us write faster, by autocompleting sentences, and write better, by suggesting alternative words, phrases, or ideas. Her work has been published in top-tier venues in natural language processing (e.g., ACL and NAACL), machine learning (e.g., NeurIPS), and human-computer interaction (e.g., CHI), and featured in various media outlets including "The Economist."

Vipul Raheja (Grammarly) is an Applied Research Scientist at Grammarly. He works on developing robust and scalable approaches centered around improving the quality of written communication, leveraging Natural Language Processing and Machine Learning. His research interests lie at the intersection of text editing and controllable text generation.

Thiemo Wambsganss (École Polytechnique Fédérale de Lausanne) is a PostDoc at the Machine Learning for Education Laboratory at the Swiss Federal Institute of Technology in Lausanne (EPFL). His work aims to leverage methods from Natural Language Processing and Machine Learning to provide users, in particular students, with intelligent writing feedback anytime and anywhere they want. In this vein, he strives to understand how humans perceive, interact, and learn with intelligent tools, such as with argumentative writing support, empathic writing support or with conversational agents in general.

3 WEBSITE

The workshop information will be shared via the following website: <https://in2writing.glitch.me/>. Concretely, the call for participation,

⁴<https://hai-gen2021.github.io/>

⁵<https://ctrlgenworkshop.github.io/>

the program, the schedule, the organizers, the program committee, and the accepted papers will be published.

4 PRE-WORKSHOP PLANS

Objective and target audience. The objective of our workshop is to create an interdisciplinary platform for researchers and industry practitioners to discuss, share, and build ideas and projects in the field of intelligent writing assistance. As a result of the workshop, we aim to write an article about the various design dimensions and characteristics of writing support systems from diverse perspectives. With our diverse team of organizers, we plan to not only promote attendance at our workshop in academic areas of ACM SIGCHI, such as in IUI, UbiComp or CSCW, but also encourage colleagues and students in other academic fields such as NLP, machine learning (ML), or information systems to participate. Moreover, with two of our workshop organizers working in the industry, we would also focus on inviting practitioners, industry researchers, and creative writers to participate in our workshop. To ensure a wide outreach, we will share and promote our call for workshop participation through mailing lists (e.g., ACM SIGCHI or ACL mailing lists), our website, and through the workshop organizers' personal networks, social media, and institutional mailing lists. We plan for 25-35 attendees to actively participate in our workshop at CHI. We believe this is a suitable size to engage in both networking and community building, but also in the active participation and discussion for our planned workshop concept.

Participant submission. In order to find suitable attendees, we will ask participants to submit a 2-page position paper on the design and evaluation of existing and future writing assistance. Our goal is to have a diverse and interdisciplinary subset of academic and industry representatives who are working on writing assistants in any form and are interested in doing so in the future. Hence, the workshop organizers will form a program committee to review and carefully select the incoming submissions. We plan to provide at least two reviews for each submission to assess the novelty, quality, and relevance of participants' content. The most novel and insightful submissions will be selected to participate in our workshop. In this way, we will control the number of participants at our workshop to avoid physical space shortages. Based on the submissions, we aim to collect ideas, design concepts, or already existing designs of writing support systems to then discuss their characteristics and design dimensions in our workshop. Before the in-person workshop takes place, we will make the accepted submissions as well as the workshop schedule publicly available on our website. One week before the workshop, we will ask the participants to familiarize themselves with the contributions, and we will distribute necessary information regarding the interactive workshop activities that we plan to engage participants in.

5 IN-PERSON FORMAT AND ASYNCHRONOUS ENGAGEMENT

We plan to organize an in-person workshop, thus requiring participants to attend in person. This is intended to foster a greater depth and quality of discussions, given that the nature of our planned activities (e.g., icebreakers, taxonomy building, and group discussion)

requires in-person collaboration. However, there will be opportunities for asynchronous engagement for anyone interested in getting involved in this area.

- **Workshop materials.** Participant submissions, recordings of keynote presentations, and the resulting write-up of the workshop will be made available on our website.
- **Twitter engagement.** Workshop organizers will post information about the workshop, including events that unfold during the day of the workshop, on Twitter with a dedicated hashtag. This will allow interested parties to engage in a conversation with attendees asynchronously.
- **Post-workshop literature review.** We will put out a call for researchers interested in participating in a formal literature review of writing support tools, based on the results of the taxonomy created during the workshop. There will be opportunities to submit systems to be included in the review and to contribute to annotations as part of this effort.

6 WORKSHOP STRUCTURE

Our workshop will consist of two parts. In the first part, to understand the landscape of writing assistants, participants will collaboratively create a taxonomy of writing assistants. In the second part, based on topical categories identified during the first part, participants will collectively discuss the current status of writing assistants and potential research agendas.

6.1 Part 1: Create a taxonomy of writing assistants

The goal of the first part is to create a taxonomy of writing assistants, including existing and future tools. The objective of building the taxonomy is to capture a spectrum of topics behind the nuances and characteristics of writing assistants. These topics may include user needs, evaluation, prototyping, pedagogical implications, ethics, and bias, as well as potential issues with NLP techniques. We hope this discussion and resulting taxonomy can serve as a guide for researchers or practitioners from HCI and NLP communities to design, develop, and research writing support systems more effectively.

The workshop will begin with participants introducing themselves and an icebreaker. Then, we will ask participants to divide into groups. Each group will try out existing writing assistants, discuss their design features, and create a taxonomy to categorize them. Tools and categories mentioned in participant submissions will be used as the starting point of the discussion. Then, all groups will put their taxonomies and writing assistant instances on a shared board. With all taxonomies, participants will collectively merge different taxonomies, ideally reaching 5–10 agreed categories. If time allows, with the moderation of organizers, participants will go through each category and formally define them, with example tools included. Then, the organizers will present finalized categories to participants.

6.2 Part 2: Discuss the status of current writing assistants and future research agendas

During the second part, leveraging the taxonomy created in the first part, participants will discuss specific topics about writing

Table 1: Proposed workshop schedule.

	Duration	Activity
Morning	09:00-09:20	Welcome: Introduce objectives, organizers, participants, and schedule, along with an icebreaker.
	09:20-10:10	Keynote: Presentation by a professional writer or an academic expert.
	10:10-10:25	Group formation 1: Split participants into groups for the following two activities.
	10:25-10:45	<i>Short break.</i>
	10:45-11:30	Explore tools: Participants try and discuss writing tools.
	11:30-12:00	Categorization: Based on the discussion, participants work on categorizing tools.
	12:00-13:00	<i>Lunch break.</i>
Afternoon	13:00-13:45	Merging categories: Participants convene around a shared board, attach tools and categories onto the board with sticky notes, collaboratively merge categories, and define categories if possible.
	13:45-14:00	Finalize categories: Organizers present resulted categories to participants.
	14:00-14:15	Group formation 2: Participants decide the category they want to discuss.
	14:15-14:30	<i>Short break.</i>
	14:30-15:15	Discussion 1: Participants discuss their first topic.
	15:15-15:30	Sharing findings 1: Participants share their findings from the discussion.
	15:30-15:45	<i>Short break.</i>
	15:45-16:30	Discussion 2: Participants discuss their second topic.
	16:30-16:45	Sharing findings 2: Participants share their findings from the discussion.
	16:45-17:00	Wrap up: summarize the workshop, plans for post workshops, and take group photos.

assistants. Participants will form groups based on the topic that they are interested in—which are categories defined during the first part. Our idea of discussion is to ground the taxonomy on socio-technical system perspectives to ensure we capture all relevant technical and social characteristics of writing support systems. With the given topic, participants will discuss how existing practices and research revolve around the topic and what future directions exist for the topic. We will provide participants with a shared board where they can put their results of discussions so that they can maintain their contexts and share the discussion results later. To facilitate discussions around various topics, participants will rotate between groups, while having at least one or two participants stay in the group so that they can let later joining participants know about the context of the discussion. After two rounds of discussion, participants in each group will present the result of the discussions to other workshop participants. Then, organizers will wrap up the workshop, presenting the summary of the workshop and plans for the post-workshop. The workshop will end with taking a group photo with all organizers and participants.

7 ACCESSIBILITY

We aim to make this workshop as accessible as possible.⁶ For all the submissions, we ask our participants to follow the *Guide to accessible submission*.⁷ Any resulting artifacts from the workshop including position papers will be made accessible following the accessibility guidelines before they become publicly available.

8 POST-WORKSHOP PLANS

All workshop participants will be invited and welcome to participate in the post-workshop activities and planning, composed of the following three main parts.

⁶Please refer to the <https://chi2023.acm.org/for-attendees/chi2023-accessibility-faq/> for general accessibility questions.

⁷<https://sigchi.org/conferences/author-resources/accessibility-guide/>

- **Disseminate the results of the workshop.** The accepted papers, videos, slides, discussion results, and releasable materials will be published on our website (<https://in2writing.glitch.me/>). Some of these materials can be put on the website before the workshop (e.g., accepted papers), and other materials will be uploaded after the workshop.
- **Keep the conversation going.** We will arrange an in-person networking event (e.g., dinner or social gathering) immediately after the workshop to continue the conversation. If we gather sufficient interests, we would also create a virtual place for people to continue interacting with each other after the workshop, such as a Slack community, Google Group, or even a ACM SIGCHI chapter group. Such virtual places have been used in other workshops, too.
- **Crystallize the thoughts and discussions.** One of the key activities of our workshop is to invite all the participants to collectively create a taxonomy for modern writing assistants. Our follow-up plan is to invite all the interested participants to co-write a literature review paper based on the created taxonomy, which will cover all the participants' systems and a more formal literature review. We will use the workshop's afternoon session to form interest groups who could structure the paper's discussion section. We will aim to consolidate and disseminate the result of the workshop in a conference, journal, or magazine article.

9 CALL FOR PARTICIPATION

Writing assistants have become increasingly ubiquitous, and these tools are likely to change the way we write. In our workshop, we, as a diverse group of researchers and practitioners interested in writing assistants, will create a taxonomy of writing assistants and discuss their desirable features and potential consequences.

To attend this in-person event, we ask participants to write a 2-page position paper on their perspectives on writing assistants, potentially responding to one of the following prompts:

- What are important dimensions to consider when designing intelligent writing assistants?
- How can we evaluate and compare writing assistants and their features in interactive settings?
- How can we design writing assistants for underrepresented languages, types of writers, and writing tasks?
- How can we make writing assistants more accessible and inclusive?
- What are the biggest ethical concerns and limitations of writing assistants, especially when leveraging AI?
- How do we address legal issues with copyright and the psychological sense of ownership of writing assistants?
- How will writing assistants change the way we write?

We invite submissions from not only researchers from the HCI and NLP communities, but also writers, educators, industry practitioners, students, and anyone interested in creating, using, and testing future writing assistant technologies. Participants will be invited based on demonstrating in-depth thinking about writing assistants. Although non-archival, invited participants can have their papers hosted on the workshop website. If you have any questions or concerns, please contact the organizers.

Submissions should follow CHI publication formats⁸ and will go through a double-blind peer review. The paper submission deadline will be February 23, 2023 (11:59 PM AOE). The timetable for the paper acceptance notification as well as the submission system will be announced on the website (<https://in2writing.glitch.me/>).

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⁸<https://chi2023.acm.org/submission-guides/chi-publication-formats/>